RISK MANAGEMENT PLAN
AND RISK REGISTER

ShopEase Online Store: Development of an E-commerce Web Application

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Risk register (live) : https://abhxym.github.io/project-management/assignment-4-risk-management/ShopEase_Risk_Register.html

GitHub Repository : https://github.com/Abhxym/project-management

ABSTRACT

This document is the risk management plan and risk register of the ShopEase online store project, which started on 1 September 2026 and is planned to go live on 27 January 2027. The plan defines how risks are identified, analysed, responded to and monitored: roles, budget, timing, risk categories, probability and impact scales, the probability-impact (P-I) matrix, risk thresholds and reporting.

20 project-specific risks were identified: 15 threats and 5 opportunities. Each risk is written as cause, event and effect, has an owner, a probability in percent, a cost and schedule impact, a P x I severity score from 1 to 25, an expected monetary value (EMV), a response strategy with a justification, a trigger, a fallback plan and a residual rating after the response. The register is built as an interactive web application in which the risks are plotted on separate P-I matrices for threats and opportunities; changing any value, or dragging a risk to another cell, recalculates the score, EMV, contingency cover and matrix at once.

Quantitative analysis shows a net EMV of Rs 97,250 before responses, which is covered by the contingency reserve of Rs 1,02,000. A decision tree justifies a backup payment gateway, and a Monte Carlo simulation of 10,000 runs over the project network shows that the planned responses raise the probability of going live on time from 13% to 57%.

Keywords: Risk Management Plan, Risk Register, Threat, Opportunity, Probability-Impact Matrix, Risk Score, Expected Monetary Value, Decision Tree, Monte Carlo Simulation, Contingency Reserve, Risk Response.

1. INTRODUCTION

1.1 Background

The charter (Assignment 1) listed seven high-level risks and the budget (Assignment 2) set a contingency reserve of Rs 1,02,000 from their EMV. Execution started on 1 September and the Kanban board and earned value control (Assignment 3) are in use. This document turns the early risk list into a full risk management process for the rest of the project.

1.2 Objectives

1. To define the risk management approach, roles, scales and thresholds.
2. To identify project-specific threats and opportunities and record them in a risk register with owners.
3. To rate each risk on a probability-impact matrix with a quantitative score and EMV.
4. To choose and justify a response for each risk and measure the residual risk.
5. To check the contingency reserve and the go-live date with quantitative analysis.

1.3 Threats and Opportunities

A risk is an uncertain event that, if it happens, affects at least one project objective. A **threat** has a negative effect (higher cost, delay, lower quality), for example the payment gateway approving the KYC late. An **opportunity** has a positive effect (lower cost, time saved, better quality), for example the cloud provider giving start-up credits. Both are managed in the same register with the same scales, but the response strategies are different: threats are avoided, transferred, mitigated or accepted, while opportunities are exploited, shared, enhanced or accepted.

2. RISK MANAGEMENT PLAN

2.1 Methodology

Step	What is done	Tools and data
1. Plan	This plan: approach, roles, scales, thresholds	Charter, baselines (Assignments 1 and 2)
2. Identify	Find threats and opportunities and write them as cause, event, effect	Workshops, assumption analysis, WBS and schedule review, board data

Step	What is done	Tools and data
3. Qualitative analysis	Rate probability and impact; score P x I; plot on the P-I matrix	Scales in 2.6, P-I matrix in 2.7
4. Quantitative analysis	EMV, contingency check, decision tree, Monte Carlo on the network	Interactive register, schedule model from Assignment 2
5. Plan responses	Choose a strategy, actions, owner, trigger, fallback; rate the residual risk	Response playbook (Section 5)
6. Implement responses	Owners carry out the actions as Kanban cards	Kanban board (Assignment 3)
7. Monitor	Review triggers, re-rate risks, close or add risks, report	Weekly status meeting, milestone reviews

Table 1: Risk management process

2.2 Roles and Responsibilities

Role	Responsibilities in risk management
Sponsor (R. Kulkarni)	Approves this plan; owns escalated risks; controls the management reserve; decides on risks above the PM authority
Project Manager (Abhay Maurya)	Owns the risk process and the register; runs risk reviews; controls the contingency reserve within the charter limits; reports top risks weekly
Risk owner	Named for each risk; watches the trigger; carries out the response; reports status and re-rates the risk
Team members	Raise new risks at any time (stand-up, sprint demos); carry out response actions as task cards
Operations Head	Owns business-side risks (staff adoption, UAT availability); member of the Change Control Board

Table 2: Roles and responsibilities

2.3 Budget

- **Contingency reserve Rs 1,02,000** (part of the cost baseline): for identified risks in this register. Used by the PM up to Rs 50,000 per risk event; above that with sponsor approval.
- **Management reserve Rs 51,000** (outside the cost baseline): for unidentified risks; released only by the sponsor through change control.
- **Response costs** (Rs 16,000 in total) are paid from the contingency reserve because they reduce identified risks.

2.4 Timing

Activity	When
Full risk identification workshop	At the start of execution and at each milestone (M1 to M6)
Risk review of top risks and triggers	Weekly status meeting (Friday)

Activity	When
Review of all medium and low risks	Every two weeks and at milestone reviews
New risk raised	Any time; rated by the PM within 2 working days
Quantitative analysis (EMV, Monte Carlo)	At each milestone and after any change to a high risk
Lessons learned on risks	At project closure (Z)

Table 3: Risk management timing

2.5 Risk Categories

Category	Typical sources in this project	Risks
Technical	Requirements, architecture, performance, security, hosting	T06, T07, T08, T12, O01
External	Payment gateway, courier, suppliers, regulators, cloud provider	T01, T02, T09, T10, T13, T15, O02, O03, O05
Organisational	Staff, owner, funding, availability of business users	T03, T05, T11, T14
Project management	Scope control, planning, estimating	T04, O04

Table 4: Risk breakdown structure (RBS)

2.6 Probability and Impact Scales

Rating	Probability	Cost impact	Schedule impact (on the activity)
5 Very high	70% or more	Rs 1,00,000 or more	20 working days or more
4 High	50% to 69%	Rs 50,000 to 99,999	10 to 19 days
3 Medium	30% to 49%	Rs 25,000 to 49,999	5 to 9 days
2 Low	10% to 29%	Rs 10,000 to 24,999	2 to 4 days
1 Very low	below 10%	below Rs 10,000	0 to 1 day

Table 5: Probability and impact scales

The impact rating is the higher of the cost and schedule ratings. The same scales are used for opportunities, where the impact is a saving or time gained. The scales are anchored to this project: Rs 1,00,000 is about 10% of the work package budget and 20 days is more than the whole schedule float on most paths.

2.7 Probability-Impact Matrix and Thresholds

The severity score is **P x I** (1 to 25). The matrix is divided into three zones, which set the required action:

Zone	Score	Threat	Opportunity
High	12 to 25	Response plan with owner within 2 days, reviewed weekly, reported to the sponsor	Exploit or share actively
Medium	6 to 11	Response plan, reviewed every two weeks	Enhance where it costs little
Low	1 to 5	Watch list, reviewed at milestones	Accept

Table 6: Score zones

Risk appetite and thresholds agreed with the sponsor: the go-live date may slip by at most 5 working days (charter authority limit), the total of identified risks must stay within the contingency reserve, and no high threat may remain on the critical path without an active response. A breach of a threshold is escalated to the sponsor.

2.8 Response Strategies

Threat strategy	Meaning	Opportunity strategy	Meaning
Avoid	Remove the cause or change the plan	Exploit	Make sure the opportunity happens
Transfer	Move the impact to a third party (contract, insurance)	Share	Work with a partner who can capture it
Mitigate	Reduce probability or impact	Enhance	Increase probability or impact
Accept	Active (fallback, contingency) or passive	Accept	Take it if it happens
Escalate	Outside the PM authority; owned by the sponsor	Escalate	Benefit is at business level; owned by the sponsor

Table 7: Response strategies

2.9 Reporting and Tracking

- The register records for each risk: ID, type, category, cause-event-effect, activity affected, owner, probability, cost and schedule impact, P, I, score, EMV, strategy, actions, justification, trigger, fallback, response cost, residual rating and status (Active, Watch, Escalated, Occurred, Closed).
- The weekly status report contains the top risks by score, risks whose trigger has fired, contingency used and the net EMV against the remaining contingency.
- When a risk occurs it becomes an issue: its status is set to Occurred, the fallback plan is started and the contingency used is recorded.
- The register is exported to CSV after each review and kept with the project files.

3. RISK IDENTIFICATION

Technique	Input	Risks found
Team workshop (1 Oct)	WBS, schedule, Kanban board	T04, T05, T06, T07, T08, T12, O01, O04

Technique	Input	Risks found
Assumption and constraint analysis	Charter assumptions (KYC by week 11, data on time, staff available)	T01, T02, T11
Interviews with sponsor, operations head and CA	Business calendar, cash flow, GST rules	T03, T13, T14
Vendor and partner review	Gateway, courier, WhatsApp, cloud, suppliers	T09, T10, O02, O03, O05
Regulatory review	DPDP Act 2023, Consumer Protection (E-Commerce) Rules	T15
Project data	Requirements finished 1 day late (Assignment 3 board)	T08 probability raised

Table 8: Identification techniques

Every risk is written in the form *"Because of (cause), (event) may happen, leading to (effect)"*. This keeps the cause, which the response works on, separate from the effect, which is used for the impact rating.

4. RISK REGISTER

Status date: 10 October 2026. T = threat, O = opportunity. Activity = schedule activity affected (Assignment 2). Owner codes: SP = Sponsor (R. Kulkarni), OH = Operations Head (M. Kulkarni), PM = Project Manager (Abhay Maurya), BA = Business Analyst, DEV = Developers, OPS = DevOps Engineer. EMV is positive for threats (expected cost) and negative for opportunities (expected saving).

ID	T/O	Risk (cause, event, effect)	Act.	Owner	Prob.	Cost impact	Days	P	I	Score	EMV
T01	T	**Payment gateway KYC approval delayed** Because the gateway needs a full set of KYC documents (GST, PAN, bank proof, website policy pages) and its review queue is long, approval may come after payment integration (L) is due to start on 30 Nov, leading to L, a critical path activity, cannot be tested with real transactions and go-live slips.	L	PM	30%	Rs 60,000	8	3	4	**12** High	Rs 18,000
T02	T	**Product data and photos from suppliers late or incomplete** Because 8 suppliers send data in different formats and some products have no photos, the product master file may not be complete when data migration (U) starts, leading to UAT runs on partial data and extra photo shoot days are needed.	U	BA	50%	Rs 40,000	5	4	3	**12** High	Rs 20,000

ID	T/O	Risk (cause, event, effect)	Act.	Owner	Prob.	Cost impact	Days	P	I	Score	EMV
T03	T	**Store staff resist the new order process** Because staff are used to WhatsApp and worry about extra work and job security, staff may avoid the admin panel or make errors after go-live, leading to order errors stay high (G7 not met) and extra training is needed.	W	OH	30%	Rs 30,000	0	3	3	**9** Medium	Rs 9,000
T04	T	**Scope creep from new feature requests** Because the owner sees competitor features during development, new features may be requested after the SRS sign-off, leading to development effort grows and the critical path moves.	K	PM	40%	Rs 80,000	10	3	4	**12** High	Rs 32,000
T05	T	**A developer leaves or is unavailable for long** Because the job market for developers is active and the team has only two developers, one developer may leave during development, leading to storefront or admin chain stops while a replacement is found and handed over.	K	PM	10%	Rs 1,20,000	15	2	5	**10** Medium	Rs 12,000
T06	T	**Site slows down or fails under launch traffic** Because the launch campaign brings a traffic peak on the first days, response times may exceed 3 seconds or the site may go down, leading to lost sales, bad reviews and emergency fixes during hypercare.	X	OPS	10%	Rs 50,000	3	2	4	**8** Medium	Rs 5,000
T07	T	**Security breach or customer data leak** Because an online store is a target for attacks and stores personal data, a vulnerability may be exploited, leading to DPDP Act penalties, loss of trust and emergency fixes.	Y	OPS	5%	Rs 1,20,000	5	1	5	**5** Low	Rs 6,000

ID	T/O	Risk (cause, event, effect)	Act.	Owner	Prob.	Cost impact	Days	P	I	Score	EMV
T08	T	**Requirements misunderstood, causing rework** Because user stories were written quickly and the requirements phase already slipped by one day, developers may build features differently from what the business expects, leading to rework in checkout and admin, delaying the critical path.	K	BA	30%	Rs 35,000	4	3	3	**9** Medium	Rs 10,500
T09	T	**Courier aggregator API change or account delay** Because the aggregator updates its API and account approval needs documents, the courier integration (N) may need extra work, leading to shipping labels and tracking are delayed.	N	DEV	20%	Rs 20,000	3	2	2	**4** Low	Rs 4,000
T10	T	**WhatsApp Business API approval delayed** Because message templates need approval from the provider, WhatsApp notifications (P3) may not be ready for go-live, leading to customers get SMS and e-mail only at launch.	P	DEV	30%	Rs 10,000	2	3	2	**6** Medium	Rs 3,000
T11	T	**Business users not available for UAT** Because January is the start of the wedding season and the shop is busy, the owner and staff may not attend UAT sessions on time, leading to UAT sign-off (M4) and go-live slip.	V	OH	30%	Rs 15,000	3	3	2	**6** Medium	Rs 4,500
T12	T	**Cloud hosting cost above plan** Because test environments and storage grow during development, monthly hosting bills may exceed the planned Rs 6,000, leading to overrun on the hosting work package.	F	OPS	20%	Rs 15,000	0	2	2	**4** Low	Rs 3,000
T13	T	**GST invoice errors found late by the CA** Because GST rules differ by product type and delivery state, the CA may reject the invoice format during review (O3), leading to rework of invoices close to testing.	O	BA	20%	Rs 15,000	2	2	2	**4** Low	Rs 3,000

ID	T/O	Risk (cause, event, effect)	Act.	Owner	Prob.	Cost impact	Days	P	I	Score	EMV
T14	T	**Owner delays vendor payments in the slow season** Because shop cash flow is low in December before the wedding season, payments to the audit firm or contractors may be delayed, leading to vendors pause work, delaying the security audit.	T	SP	15%	Rs 20,000	5	2	3	**6** Medium	Rs 3,000
T15	T	**New data protection rules need consent changes** Because rules under the DPDP Act 2023 are still being notified, new consent or notice requirements may apply before go-live, leading to changes to registration and privacy screens.	H	PM	15%	Rs 25,000	3	2	3	**6** Medium	Rs 3,750
O01	O	**Reuse an open-source UI component library** Because a mature component library matches the approved design, the storefront can be built from ready components, leading to catalogue and checkout pages are built faster.	I	DEV	50%	Rs 30,000	4	4	3	**12** High	-Rs 15,000
O02	O	**Cloud provider start-up credits** Because the cloud provider has a programme for small Indian businesses, ShopEase may receive hosting credits, leading to hosting cost during the project falls.	F	OPS	40%	Rs 25,000	0	3	3	**9** Medium	-Rs 10,000
O03	O	**Suppliers provide professional product photos** Because two large suppliers already have catalogue photos, they may share them for ShopEase products, leading to fewer products need the paid photo shoot.	U	BA	30%	Rs 20,000	2	3	2	**6** Medium	-Rs 6,000
O04	O	**Shorter UAT through sprint demos** Because the operations head already reviews features in sprint demos, many UAT scripts may be accepted before formal UAT, leading to UAT takes fewer days.	V	PM	40%	Rs 10,000	2	3	2	**6** Medium	-Rs 4,000

ID	T/O	Risk (cause, event, effect)	Act.	Owner	Prob.	Cost impact	Days	P	I	Score	EMV
O05	O	**Courier aggregator provides an integration engineer** Because the aggregator offers integration support to merchants who sign a 12-month contract, its engineer may help build the courier integration, leading to less developer effort on N.	N	SP	30%	Rs 15,000	2	3	2	**6** Medium	-Rs 4,500

Table 9: Risk register (before response)

5. RISK RESPONSE PLAYBOOK

The response for each risk was chosen by comparing the cost of the response with the reduction in EMV and by the zone of the risk. High threats on the critical path (T01, T04) get active responses even if they cost money; low risks with an EMV below about Rs 5,000 are accepted with a fallback plan, because a response would cost more than the expected loss. Escalation is used only when the cause is outside the project manager's authority (T14).

ID	Strategy	Response actions	Justification	Trigger	Fallback	Resp. cost	Residual P x I
T01	**Mitigate**	KYC documents submitted on 08 Oct; weekly follow-up; build and test in sandbox; open a second gateway account in parallel (decision tree, Section 6.3)	Response cost Rs 8,000 is lower than the EMV reduction of Rs 16,500, and it protects the critical path.	No approval by 13 Nov (week 11)	Go live with the second gateway; switch later	Rs 8,000	2 x 2 = **4**
T02	**Mitigate**	Data template sent with supplier calls in week 3; fortnightly completeness check; photo shoot for missing items already in budget	No extra cost; U has 12 days of float, so the residual schedule effect is absorbed.	Less than 60% of SKUs complete by 16 Nov	Launch with products that are complete; add the rest in hypercare	-	3 x 3 = **9**
T03	**Mitigate**	Two staff in requirements and UAT; demos in the shop every sprint; Marathi manual; show time saved	Rs 5,000 of extra demo time reduces EMV by Rs 6,000 and protects the business goal G7.	Negative feedback in sprint demos	Extra hands-on sessions during hypercare	Rs 5,000	2 x 2 = **4**
T04	**Avoid**	Formal change control (CCB); MoSCoW priorities; Phase 2 backlog for new ideas; changes over 2 person-days go to the CCB	Change control removes the uncontrolled part of the risk at no cost; only approved, funded changes remain.	Change request raised	Trade a Could-have feature for the new one	-	2 x 3 = **6**

ID	Strategy	Response actions	Justification	Trigger	Fallback	Resp. cost	Residual P x I
T05	**Transfer**	Contract with the staffing firm requires a replacement within 5 days at their cost; code reviews and shared modules so both developers know all code	Most of the cost moves to the staffing firm by contract; knowledge sharing lowers the schedule effect.	Resignation or leave over 5 days	Contract developer from the staffing firm	-	2 x 3 = **6**
T06	**Mitigate**	Load test with 500 users (T1); auto-scaling and caching; campaign starts after 2 stable days	Load testing is already in the plan; the residual risk is low.	Load test fails the 3-second target	Add servers for launch week (Rs 6,000)	-	1 x 2 = **2**
T07	**Mitigate**	No card data stored (gateway); firewall and hardening (R); external VAPT (T2); least-privilege admin accounts; daily backups	Controls are already budgeted in R and T; impact is very high so the risk is kept on the watch list.	VAPT finds a critical issue	Incident response plan; take site offline and restore	-	1 x 4 = **4**
T08	**Mitigate**	Prototype test with 5 customers (D4); acceptance criteria on every card; demo to operations head at the end of each sprint	Small review cost; sprint demos catch problems early when rework is cheap.	More than 2 change requests from a sprint demo	Use contingency for rework days	Rs 3,000	2 x 2 = **4**
T09	**Mitigate**	Account opened by week 13; integrate through the aggregator SDK; keep manual label printing as fallback	No extra cost; manual fallback keeps go-live safe.	No API keys by 23 Nov	Manual labels for the first month	-	2 x 2 = **4**
T10	**Accept**	Active acceptance: launch with SMS and e-mail; add WhatsApp when approved	Impact is small and any response would cost more than the EMV of Rs 3,000.	Templates not approved by 16 Dec	SMS and e-mail only	-	3 x 2 = **6**
T11	**Mitigate**	UAT dates (13 to 20 Jan) booked in the shop calendar now; sessions in the morning before opening	Booking dates early costs nothing and removes most of the risk.	UAT session cancelled	Evening sessions; sponsor nominates a deputy	-	2 x 2 = **4**
T12	**Accept**	Budget alerts at 80% of the monthly plan; switch off test servers at night	Low EMV; simple cost controls are enough.	Budget alert	Use contingency	-	2 x 1 = **2**
T13	**Mitigate**	CA consulted on HSN codes and rates in requirements; sample invoices reviewed as soon as O1 is done	Early review costs nothing extra.	CA review finds an error	Fix during testing using developer support days	-	2 x 1 = **2**
T14	**Escalate**	Escalated to the sponsor: business-level cash planning is outside the PM authority; payment dates agreed in vendor contracts	The cause is the business cash flow, which only the sponsor controls.	Invoice unpaid after 15 days	Sponsor releases management reserve	-	2 x 2 = **4**

ID	Strategy	Response actions	Justification	Trigger	Fallback	Resp. cost	Residual P x I
T15	**Accept**	Watch notifications monthly; consent screens built as configurable text	Outside project control; a configurable design lowers the impact at no extra cost.	New rule notified	Change request to the CCB	-	2 x 2 = **4**
O01	**Exploit**	Library chosen in the architecture (E); designer maps mock-ups to library components	Exploit makes the opportunity almost certain at no cost; saves effort on the critical path.	Architecture review	-	-	5 x 3 = **15**
O02	**Enhance**	Application with GST and company documents submitted; follow up with the provider account manager	Small effort raises the probability of a Rs 25,000 saving.	Credits approved	-	-	4 x 3 = **12**
O03	**Enhance**	Request photos from all suppliers with the data template; offer supplier branding on product pages	Asking early increases the chance at no cost.	Supplier sends photo set	-	-	4 x 2 = **8**
O04	**Enhance**	Record demo acceptance against UAT scripts; formal UAT covers only what is left	Uses meetings that already happen; may give schedule reserve before go-live.	Over 50% of scripts accepted in demos	-	-	4 x 2 = **8**
O05	**Share**	Sponsor negotiates the 12-month contract including integration support; benefit shared with the aggregator	Sharing with a partner who has the skills; the contract is needed anyway.	Contract signed	-	-	4 x 2 = **8**

Table 10: Response playbook with residual ratings

5.1 Strategy Mix

Strategy	Risks
Avoid	T04
Transfer	T05
Mitigate	T01, T02, T03, T06, T07, T08, T09, T11, T13
Accept	T10, T12, T15
Escalate	T14
Exploit	O01
Share	O05
Enhance	O02, O03, O04

Table 11: Risks by response strategy

6. RISK ANALYSIS

6.1 Probability-Impact Mapping

The interactive register plots threats and opportunities on separate 5 x 5 matrices. Before the responses, 3 threats are in the high zone: T01 (payment gateway kyc approval delayed), T02 (product data and photos from suppliers late or incomplete), T04 (scope creep from new feature requests). After the responses no threat remains in the high zone, while the opportunities move up because they are exploited or enhanced.



Figure 1: P-I matrices before response (interactive register)



Figure 2: P-I matrices after response

Rank	ID	Risk	Owner	Score	EMV
1	**T04**	Scope creep from new feature requests	PM	12	Rs 32,000
2	**T02**	Product data and photos from suppliers late or incomplete	BA	12	Rs 20,000

Rank	ID	Risk	Owner	Score	EMV
3	**T01**	Payment gateway KYC approval delayed	PM	12	Rs 18,000
4	**O01**	Reuse an open-source UI component library	DEV	12	-Rs 15,000
5	**T05**	A developer leaves or is unavailable for long	PM	10	Rs 12,000
6	**T08**	Requirements misunderstood, causing rework	BA	9	Rs 10,500
7	**O02**	Cloud provider start-up credits	OPS	9	-Rs 10,000
8	**T03**	Store staff resist the new order process	OH	9	Rs 9,000
9	**T06**	Site slows down or fails under launch traffic	OPS	8	Rs 5,000
10	**O03**	Suppliers provide professional product photos	BA	6	-Rs 6,000

Table 12: Top 10 risks by score (then by EMV)

6.2 Expected Monetary Value and Contingency

	Before response	After response
Threat EMV	Rs 1,36,750	Rs 35,850
Opportunity EMV (saving)	-Rs 39,500	-Rs 67,000
Net EMV	**Rs 97,250**	**-Rs 31,150**
Response cost	-	Rs 16,000
Contingency reserve	Rs 1,02,000	Rs 1,02,000
Reserve left after net EMV (and response cost)	Rs 4,750	Rs 1,17,150

Table 13: EMV and contingency reconciliation

Before any response the net EMV (Rs 97,250) is just inside the contingency reserve (Rs 1,02,000), so the reserve set in Assignment 2 is still adequate even with the eight new threats, because the opportunities offset part of them. The planned responses cost Rs 16,000 and reduce the threat EMV by Rs 1,00,900, which is a strong return on the response cost.

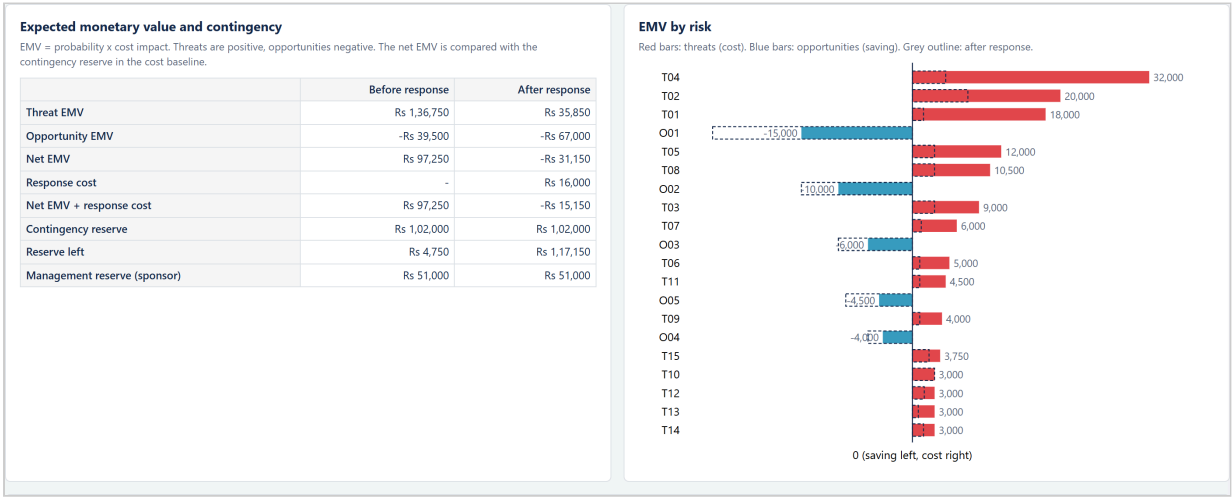


Figure 3: EMV and contingency panel of the interactive register; bars show EMV per risk before (filled) and after (dashed) response

6.3 Decision Tree: Backup Payment Gateway (T01)

The highest-scoring threat on the critical path is the payment gateway KYC delay. The decision is whether to open a second gateway account now (cost Rs 8,000 for documents and a basic integration) so that a delay would cost only Rs 15,000 instead of Rs 60,000.

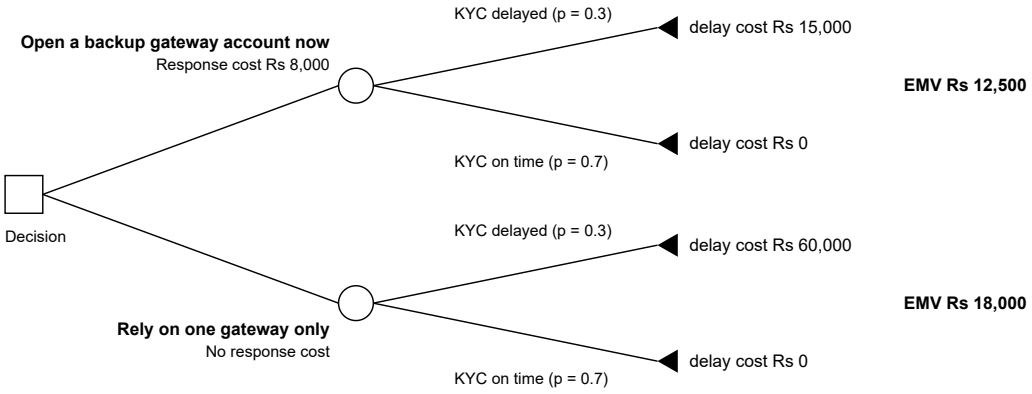


Figure 4: Decision tree for the backup payment gateway

Option	Calculation	EMV
Backup gateway now	Rs 8,000 + 0.3 x Rs 15,000	**Rs 12,500**
One gateway only	0.3 x Rs 60,000	Rs 18,000

Table 14: Decision tree result

The backup option has the lower expected cost (Rs 12,500 against Rs 18,000), saving Rs 5,500 on average, and it also limits the worst case on the critical path. The backup gateway was therefore chosen as the response to T01.

6.4 Monte Carlo Simulation

The interactive register contains a Monte Carlo simulation of the project network from Assignment 2. In each run, every activity duration is drawn from its PERT (beta) distribution, each risk occurs with its probability and adds (threat) or removes (opportunity) 50% to 150% of its days on its activity and its cost, and the forward pass is recalculated with all FS, SS and FF links and lags. Results of 10,000 runs (seed 2026):

Scenario	On-time go-live (by 27 Jan 2027)	P50 go-live	P80 go-live	Average delay	P80 cost	Within cost baseline
Duration uncertainty only	26%	29 Jan 2027	02 Feb 2027	2.0 days	Rs 10,17,800	100%
With risks, before response	13%	09 Feb 2027	19 Feb 2027	9.8 days	Rs 11,78,118	56%

Scenario	On-time go-live (by 27 Jan 2027)	P50 go-live	P80 go-live	Average delay	P80 cost	Within cost baseline
With risks, after response	57%	27 Jan 2027	01 Feb 2027	0.1 days	Rs 10,12,947	100%

Table 15: Monte Carlo results (10,000 runs)

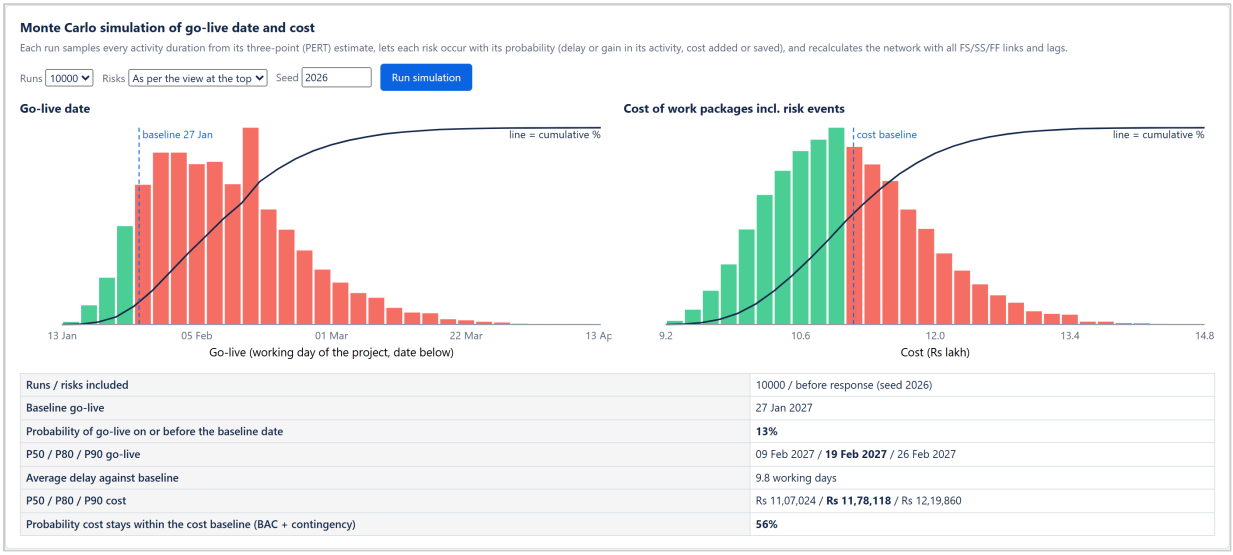


Figure 5: Monte Carlo panel before response: go-live and cost distributions with cumulative curves

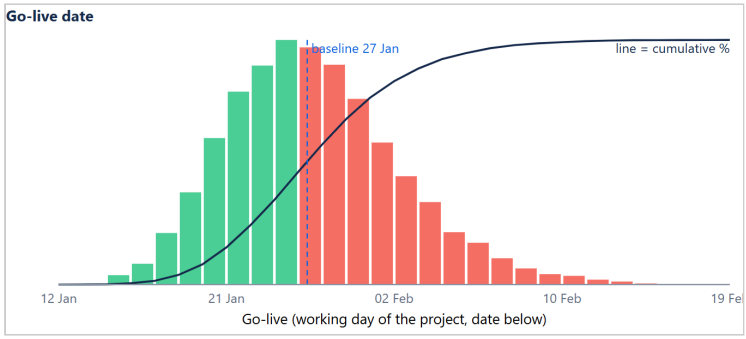


Figure 6: Go-live distribution after response

- Even without risk events the baseline date has only a 26% chance, because the PERT distributions are skewed to the right and several paths merge before testing (merge bias). The baseline is a most-likely plan, not a safe one.
- With the identified risks and no responses, the chance falls to 13% and the P80 date is 19 Feb 2027, well beyond the 5-day tolerance.
- With the planned responses the chance rises to 57% and the P80 date is 01 Feb 2027, within the 5-day tolerance. The cost stays within the cost baseline in 100% of the runs.
- To reach a higher confidence, the crashing option from Assignment 2 (requirements and design, 5 days for Rs 27,000) is kept ready as a schedule

reserve.

7. RISK STATUS ON 10 OCTOBER 2026

ID	Status	Update
T01	Active	KYC documents submitted to the gateway on 08 Oct; backup gateway account application started; sandbox integration planned in K
T02	Active	Data template sent to all 8 suppliers; 3 have replied; first completeness check on 16 Oct
T04	Active	Change control in use; one request (festival banner) handled inside existing scope, no CCB needed
T05	Active	Replacement clause added to the staffing contract on 07 Oct (transfer in place)
T08	Active	Requirements finished 1 day late; probability kept at 30% and prototype test D4 brought forward
T14	Escalated	Sponsor agreed vendor payment dates for December; reviewed at the next monthly meeting
O01	Active	Component library selected in the architecture document (E1), designer mapping started
O02	Active	Start-up credit application submitted on 09 Oct

Table 16: Status of key risks at the review

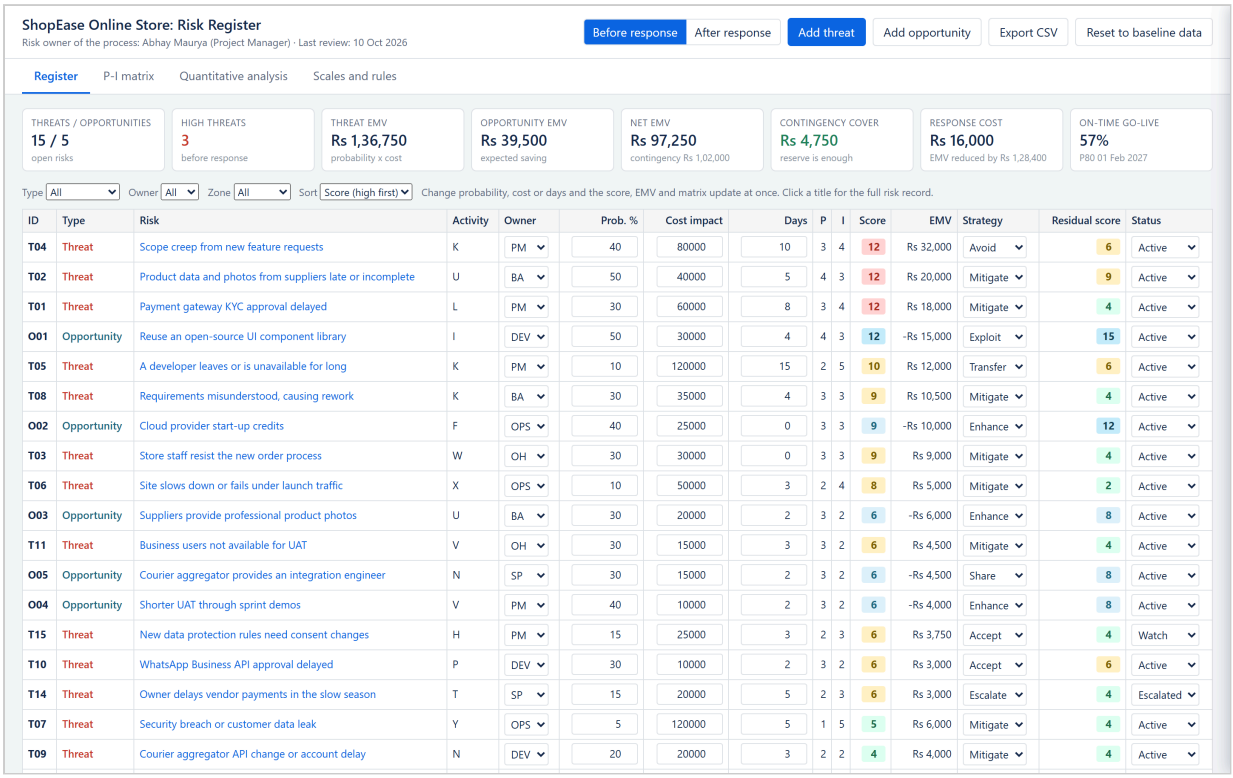
8. THE INTERACTIVE RISK REGISTER

The register is a single HTML file (ShopEase_Risk_Register.html) that runs offline in any browser. It is published with GitHub Pages and can be opened directly at https://abhxym.github.io/project-management/assignment-4-risk-management/ShopEase_Risk_Register.html; the source is in the assignment-4-risk-management folder of the GitHub repository. Its tabs are:

- **Register:** all risks with editable owner, probability, cost, days, strategy and status; the P, I, score and EMV are recalculated as soon as a value changes; filters by type, owner and zone; sorting by score or EMV; a full record for each risk with cause, event, effect, response, trigger, fallback and residual values; adding and deleting risks; CSV export.
- **P-I matrix:** separate matrices for threats and opportunities with risk IDs in their cells; dragging a risk to another cell re-rates it; switch between before and after response.
- **Quantitative analysis:** EMV totals, contingency cover, EMV chart per risk and the Monte Carlo simulation of go-live date and cost.
- **Scales and rules:** the scales, zones and strategies of this plan.

Example of the dynamic behaviour: raising the probability of T01 from 30% to 60% in the register moved it from P3 x I4 = 12 to P4 x I4 = 16, its EMV from Rs 18,000 to

Rs 36,000, and the net EMV from Rs 97,250 to Rs 1,15,250, which is above the contingency reserve. The contingency cover indicator turned red, which is the signal to escalate to the sponsor.



ShopEase Online Store: Risk Register

Risk owner of the process: Abhay Maurya (Project Manager) · Last review: 10 Oct 2026

Before response | After response | Add threat | Add opportunity | Export CSV | Reset to baseline data

Register | P-I matrix | Quantitative analysis | Scales and rules

THREATS / OPPORTUNITIES	HIGH THREATS	THREAT EMV	OPPORTUNITY EMV	NET EMV	CONTINGENCY COVER	RESPONSE COST	ON-TIME GO-LIVE
15 / 5	3	Rs 1,36,750	Rs 39,500	Rs 97,250	Rs 4,750	Rs 16,000	57%
open risks	before response	probability x cost	expected saving	contingency Rs 1,02,000	reserve is enough	EMV reduced by Rs 1,28,400	P80 01 Feb 2027

Type All | Owner All | Zone All | Sort Score (high first) | Change probability, cost or days and the score, EMV and matrix update at once. Click a title for the full risk record.

ID	Type	Risk	Activity	Owner	Prob. %	Cost impact	Days	P	I	Score	EMV	Strategy	Residual score	Status
T04	Threat	Scope creep from new feature requests	K	PM	40	80000	10	3	4	12	Rs 32,000	Avoid	6	Active
T02	Threat	Product data and photos from suppliers late or incomplete	U	BA	50	40000	5	4	3	12	Rs 20,000	Mitigate	9	Active
T01	Threat	Payment gateway KYC approval delayed	L	PM	30	60000	8	3	4	12	Rs 18,000	Mitigate	4	Active
O01	Opportunity	Reuse an open-source UI component library	I	DEV	50	30000	4	4	3	12	-Rs 15,000	Exploit	15	Active
T05	Threat	A developer leaves or is unavailable for long	K	PM	10	120000	15	2	5	10	Rs 12,000	Transfer	6	Active
T08	Threat	Requirements misunderstood, causing rework	K	BA	30	35000	4	3	3	9	Rs 10,500	Mitigate	4	Active
O02	Opportunity	Cloud provider start-up credits	F	OPS	40	25000	0	3	3	9	-Rs 10,000	Enhance	12	Active
T03	Threat	Store staff resist the new order process	W	OH	30	30000	0	3	3	9	Rs 9,000	Mitigate	4	Active
T06	Threat	Site slows down or fails under launch traffic	X	OPS	10	50000	3	2	4	8	Rs 5,000	Mitigate	2	Active
O03	Opportunity	Suppliers provide professional product photos	U	BA	30	20000	2	3	2	6	-Rs 6,000	Enhance	8	Active
T11	Threat	Business users not available for UAT	V	OH	30	15000	3	3	2	6	Rs 4,500	Mitigate	4	Active
O05	Opportunity	Courier aggregator provides an integration engineer	N	SP	30	15000	2	3	2	6	-Rs 4,500	Share	8	Active
O04	Opportunity	Shorter UAT through sprint demos	V	PM	40	10000	2	3	2	6	-Rs 4,000	Enhance	8	Active
T15	Threat	New data protection rules need consent changes	H	PM	15	25000	3	2	3	6	Rs 3,750	Accept	4	Watch
T10	Threat	WhatsApp Business API approval delayed	P	DEV	30	10000	2	3	2	6	Rs 3,000	Accept	6	Active
T14	Threat	Owner delays vendor payments in the slow season	T	SP	15	20000	5	2	3	6	Rs 3,000	Escalate	4	Escalated
T07	Threat	Security breach or customer data leak	Y	OPS	5	120000	5	1	5	5	Rs 6,000	Mitigate	4	Active
T09	Threat	Courier aggregator API change or account delay	N	DEV	20	20000	3	2	2	4	Rs 4,000	Mitigate	4	Active

Figure 7: Register tab of the interactive risk register

9. CONCLUSION

The risk management plan sets clear scales, thresholds, roles and a review rhythm for the ShopEase project. The register contains 15 project-specific threats and 5 opportunities, each with an owner, a quantitative P x I score and EMV, and a response that is justified by cost and by its effect on the critical path. Quantitative analysis confirms that the contingency reserve of Rs 1,02,000 covers the net EMV of the identified risks, that a backup payment gateway is worth its cost, and that the planned responses raise the chance of an on-time go-live from 13% to 57%. Because the register is interactive, it is used in every weekly review to re-rate risks and see their effect on the P-I matrix, the reserve and the go-live date at once.

❖ GitHub Repository

All four Project Management assignments of the ShopEase project (reports, Excel model, Kanban board and risk register) are available on GitHub:

GitHub Repository Link:
https://github.com/Abhxym/project-management

Risk register (opens in the browser):
https://abhxym.github.io/project-management/assignment-4-risk-management/ShopEase_Risk_Register.html

Files for this assignment (assignment-4-risk-management):

- ShopEase_Risk_Register.html - the interactive risk register with P-I matrices, EMV and Monte Carlo simulation.
- PM_Assignment4_Risk_Management_Plan.pdf and .docx - this report.

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